



LAUNDRY

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Class : 1CC4

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PROJECT ON

LAUNDRY

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LAUNDRY

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Date of Submission: 20/10/2025

CERTIFICATE

This is to certify that this report titled “Laundry” embodies the original work done by Muhammad Alfin Mulya, Muhammad Akasyah, Alezuna M. Nadhif Pohan. Project in partial fulfillment of their course requirement at CEP-CCIT FTUI.

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Praise and gratitude are always offered to God Almighty for the gift of knowledge and the opportunity given to us, allowing us to complete this paper to the best of our ability.

Laundry management is a service business that requires efficiency in transaction processing and accuracy in calculating laundry costs. In this project, we developed a CLI-based Laundry Program, where users can easily select their desired laundry package and input the weight of their clothes. The program will then automatically calculate the total cost based on the selected package and weight.

With the help of technology, especially programming, daily service processes such as recording orders, calculating costs, and managing transactions can be simplified and optimized. This program is designed to make laundry management more efficient, accurate, and user-friendly.

Through this project, we aim to apply fundamental programming concepts to solve real-world problems and demonstrate how a simple command-line interface can provide practical solutions for small-scale service businesses like laundry operations.

We hope this work enhances our understanding of how technology can be implemented in everyday business processes and inspires further innovation.

BACKGROUND

Technology continues to advance rapidly, pushing businesses in all sectors to adopt digital systems that improve accuracy and efficiency. In the service industry, especially in laundry businesses, the ability to process transactions quickly and accurately is essential to maintain customer satisfaction. Laundry Mama Tangguh, as a service provider, still uses manual transaction methods such as handwritten notes, which often lead to recording errors and slow data processing. This not only affects operational efficiency but can also reduce customer trust due to delays and inconsistent pricing records.

To address these issues, our group (Group 10) developed a Laundry Cashier System using the C programming language, designed to operate through a Command Line Interface (CLI). This system helps automate the process of recording laundry transactions and calculating service costs based on package type and laundry weight. Users can also view transaction history directly without the need for manual bookkeeping.

Through this system, Laundry Mama Tangguh can reduce manual errors, speed up transaction handling, and ensure that customer data is processed in a more organized way. Although this version still stores data temporarily in memory and does not yet include permanent storage or receipt printing features, it effectively demonstrates the basic concept of digital transaction management. In the future, this system can be further developed to include database integration and automated reporting features to fully support the company's operational workflow.

SYSTEM ANALYSIS

The system developed is a Laundry Transaction Management Program built using the C programming language and operates through a Command Line Interface (CLI). This application helps manage customer laundry transactions efficiently by providing structured menus, automated price calculations, and transaction history tracking.

Design & Technical:

1. uses a main menu (Laundry, View History, Exit).
2. transaction data stored in a struct Transaksi(name, package, weight, total).
3. uses if-else, switch-case, and do-while for flow control and price calculation.
4. price calculated automatically based on package and weight.
5. all data stored temporarily in memory (no database)

Objectives:

- simplify and speed up transactions,
- reduce manual calculation errors,
- provide a simple and easy-to-use interface.

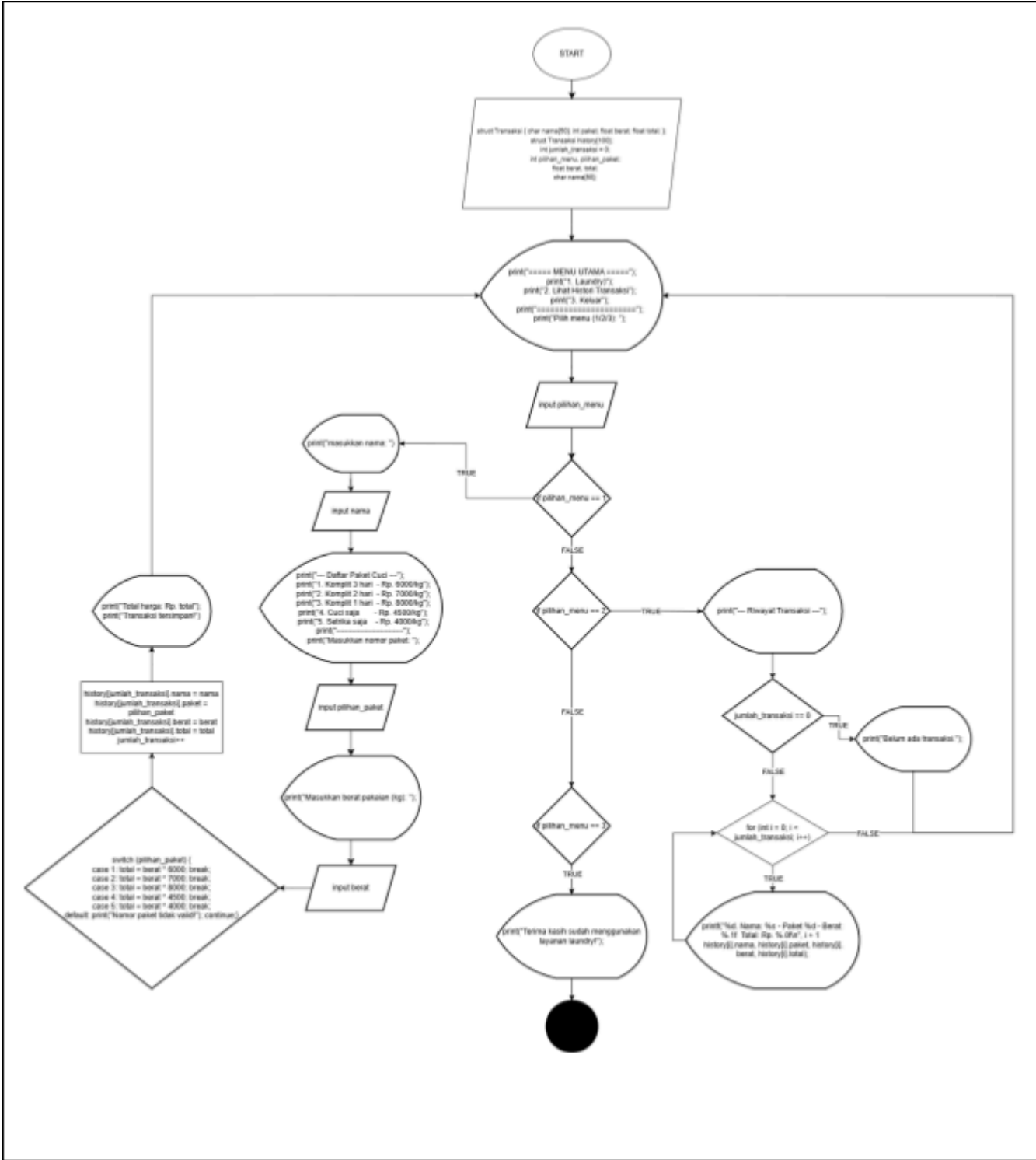
Limitations:

- data disappears after program ends,
- no receipt printing yet,
- no login or permanent storage system.

Conclusion:

even though it's simple, this system already shows the core process of managing laundry transactions efficiently and in a structured way.

FLOWCHART



SOURCE CODE

```
#include <stdio.h>
#include <string.h>

struct Transaksi {
    char nama[50];
    int paket;
    float berat;
    float total;
};

int main() {
    struct Transaksi history[100];
    int jumlah_transaksi = 0;

    int pilihan_menu, pilihan_paket;
    float berat, total;
    char nama[50];

    do {
        printf("\n===== MENU UTAMA =====\n");
        printf("1. Laundry\n");
        printf("2. Lihat Histori Transaksi\n");
        printf("3. Keluar\n");
        printf("=====\n");
        printf("Pilih menu (1/2/3): ");
        scanf("%d", &pilihan_menu);

        if (pilihan_menu == 1) {
            printf("\n--- Daftar Paket Cuci ---\n");
            printf("1. Komplit 3 hari - Rp. 6000/kg\n");
            printf("2. Komplit 2 hari - Rp. 7000/kg\n");
            printf("3. Komplit 1 hari - Rp. 8000/kg\n");
            printf("4. Cuci saja - Rp. 4500/kg\n");
            printf("5. Setrika saja - Rp. 4000/kg\n");
            printf("-----\n");
```

```

printf("\nMasukkan nama pelanggan: ");
scanf(" %[^\\n]", nama);
printf("Masukkan nomor paket: ");
scanf("%d", &pilihan_paket);
printf("Masukkan berat pakaian (kg): ");
scanf("%f", &berat);

switch (pilihan_paket) {
    case 1: total = berat * 6000; break;
    case 2: total = berat * 7000; break;
    case 3: total = berat * 8000; break;
    case 4: total = berat * 4500; break;
    case 5: total = berat * 4000; break;
    default:
        printf("Nomor paket tidak valid!\\n");
        continue;
}

printf("Total harga: Rp. %.0f\\n", total);

strcpy(history[jumlah_transaksi].nama, nama);
history[jumlah_transaksi].paket = pilihan_paket;
history[jumlah_transaksi].berat = berat;
history[jumlah_transaksi].total = total;
jumlah_transaksi++;

printf("Transaksi atas nama %s berhasil disimpan!\\n", nama);

} else if (pilihan_menu == 2) {
    printf("\\n--- Riwayat Transaksi ---\\n");
    if (jumlah_transaksi == 0) {
        printf("Belum ada transaksi.\\n");
    } else {
        for (int i = 0; i < jumlah_transaksi; i++) {
            printf("%d. Nama: %s - Paket %d - Berat: %.1f kg - Total: Rp. %.0f\\n",
                i + 1,

```

```
        history[i].nama,  
        history[i].paket,  
        history[i].berat,  
        history[i].total);  
    }  
}  
  
} else if (pilihan_menu == 3) {  
    printf("Terima kasih sudah menggunakan layanan laundry!\n");  
    break;  
  
} else {  
    printf("Menu tidak valid!\n");  
}  
  
} while (1);  
  
return 0;  
}
```

OUTPUT PROGRAM

When the program is executed, it will first display the main menu (MENU UTAMA) on the screen.

This main menu provides several options that the user can choose.

1. When choosing option 1 (Laundry)
The program displays a complete list of available laundry packages along with their prices per kilogram.
2. Then, the user is asked to input the name of the person who wants to do laundry, package number and the weight of the clothes.
3. After the data is entered, the program automatically calculates the total price based on the selected package and displays the result on the screen.
4. The transaction is then saved into the transaction history.
5. When choosing option 2 (View Transaction History)
The program will display all previous transactions under the same user name, including the package number, laundry weight, and total payment.
6. If there are no previous transactions, the program will show the message “Belum ada transaksi.”
7. When choosing option 3 (Exit)
The program displays a thank-you message and ends the execution.
8. When entering an invalid option
If the user enters a number other than 1, 2, or 3, the program displays the message “Menu tidak valid!” and automatically returns to the main menu.

The program continues to display the main menu repeatedly because it uses the `do { ... } while(1);` looping structure, which keeps the program running until the user chooses to exit.